



Bilimni
baholash
agentligi



BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA
Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA

FANIDAN

TEST TOPSHIRIQLARI KITOB

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **Shohruh Matematika jamoasi**

Telegram bot: [@bm_rashtest_bot](https://t.me/bm_rashtest_bot)

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

- yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.
- ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'inishda yozing.
- moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!

SHOHRUH MOCK-2. 30.09.2026

1. a va b butun sonlar, $a + b = 10$ va $1 \leq a < 5$ bo'lsa, b ning qabul qilishi mumkin bo'lgan eng kichik va eng katta qiymatlari yig'indisini toping.

A) 12
B) 13
C) 14
D) 15

2. Hisoblang:

$$0! + \frac{1!}{0!} + \frac{2!}{1!} + \frac{3!}{2!} + \frac{4!}{3!} + \cdots + \frac{10!}{9!}.$$

A) 45
B) 56
C) 55
D) 72

3. Bir kun, Anvar A shahardan B shaharga 60 km/h bilan yo'lga chiqdi. Bir soatdan keyin, mashina buzildi, va Anvar darrov Boburni yordamga chaqirdi. Bobur A dan bir xil yo'l bo'ylab 80 km/h tezlik bilan yo'lga chiqdi. Bobur Anvar bilan ko'rishganidan keyin, u Anvarning mashinasini B gacha tortib bordi, 40 km/h tezlikda. A va B orasidagi masofa 180 km. Anvar bu sayohatga qancha vaqt sarfladi? (soat)

A) 4, 5
B) 4, 25
C) 4, 75
D) 5

4. Jasur o'zidagi so'm ko'rinishidagi pulining 80% iga kumush sotib oldi va bir oz vaqtdan keyin uni sotib, bu savdodan 10% zarar ko'rdi. So'ngra yonida qolgan barcha puliga oltin sotib olib, uni ham bir oz vaqt o'tib sotdi. Oltin oldi-sotdisidan 25% foyda ko'rgan bo'lsa, Jasurning oxirgi holatdagi puli dastlabki puliga nisbatan necha foizga ortgan?

A) 12%
B) 15%
C) 16%
D) 18%

5. Hisoblang:

$$\frac{2^{-2} \cdot 5^3 \cdot 10^{-4}}{2^{-3} \cdot 5^2 \cdot 10^{-5}}.$$

A) 100
B) 10
C) 1000
D) 50

6. Hisoblang:

$$\sqrt{11^2 + 60^2} + \sqrt{13^2 + 84^2} + \sqrt{15^2 + 112^2} + \sqrt{17^2 + 144^2}.$$

- A) 402
- B) 400
- C) 404
- D) 406

7. Ushbu tenglikdan a ning qiymatini toping:

$$\frac{3\sqrt{7}}{\sqrt{16 + 6\sqrt{7}} - \sqrt{16 - 6\sqrt{7}}} = a - \frac{7}{2}.$$

- A) -5
- B) 4
- C) -4
- D) 5

8. Agar ushbu $\log_{12} 1250$, $\log_{12} x$, $\log_{12} y$, $\log_{12} 162$ sonlari ketma-ketligi arifmetik progressiyani tashkil etsa, x ni toping.

- A) $150\sqrt[3]{75}$
- B) $15\sqrt[3]{75}$
- C) $50\sqrt[3]{75}$
- D) 54

9. Cheksiz kamayuvchi geometrik progressiya dastlabki 3 ta hadining yig'indisi 91 ga teng. Agar shu sonlarga mos ravishda 25, 27 va 1 sonlari qo'shilsa, arifmetik progressiya hosil bo'ladi. Geometrik progressiyaning 7-hadini toping.

- A) $\frac{1}{9}$
- B) 5103
- C) $\frac{7}{81}$
- D) 5100

10. $x = \sqrt[3]{5} + 1$ bo'lsa, $x^3 - 3x^2 + 3x$ ning qiymatini toping.

- A) 4
- B) 6
- C) 5
- D) 7

11. Amallarni bajaring:

$$\left[\frac{y^2 + z^2 - x^2}{2yz} + 1 \right] \cdot \left[\frac{(x + z - y)(x + y - z)}{(x + y + z)(y + z - x)} + 1 \right].$$

- A) 1
- B) 1,5

C) 2

D) 3

12. $a = \sin x$, $b = \cos x$ bo'lsa quyidagini soddalashtiring:

$$\frac{a^3 - ba^2 - b^3 + ab^2}{a - b}$$

A) a^2 B) -1 C) $-b^2$

D) 1

13. Tenglamaning $[0; 2\pi]$ kesmadagi ildizlari sonini aniqlang:

$$1 + \cos 2x - 2 \sin^2 x = 1 .$$

A) 4 ta

B) 3 ta

C) 2 ta

D) 5 ta

14. Tenglamaning ildizlari yig'indisini (Agar u bitta bo'lsa, shu ildizini) toping:

$$2^{2x+1} - 5 \cdot 6^x + 3^{2x+1} = 0 .$$

A) 2

B) -1

C) 0

D) -4

15. Tengsizlikning eng katta va eng kichik butun yechimlari yig'indisini toping:

$$(x^2 - 2x - 24) \cdot \sqrt{10x - x^2} < 0 .$$

A) 3

B) 4

C) 6

D) 7

16. Quyidagi tenglamalar sistemasini qanoatlantiruvchi x va y qiymatlarining ko'paytmasi nechaga teng:

$$\begin{cases} x^2 - x - 1 = y \\ x + \sqrt{y - 1} = 5 \end{cases} .$$

A) 6

B) 10

C) 15

D) 12

17. Tengsizlikni yeching:

$$(70^x - 71^x) \cdot \log_x(x^2 - 7x + 13) > 0 .$$

- A) $(1; 3)$
- B) $(0; 1) \cup (3; 4)$
- C) $(0; 1) \cup (4; \infty)$
- D) $(0; 1) \cup (3; \infty)$

18. Quyidagi tenglamani qanoatlantiruvchi turli x qiymatlarining yig'indisi nechaga teng?

$$|4x^2 - 4x + 1| - |4x - 2| = 0 .$$

- A) 1,5
- B) 1
- C) 2
- D) 2,25

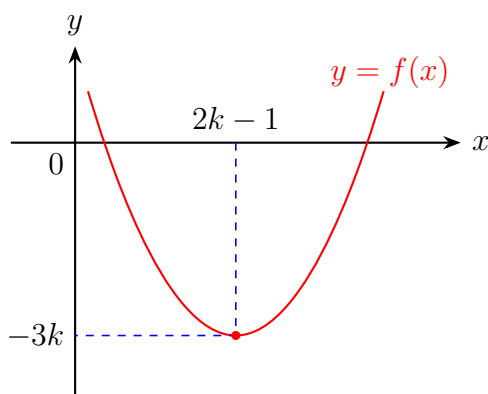
19. $2 \cdot x^2 + (2 \cdot \cos \alpha) \cdot x - \sin^2 \alpha = 0$ tenglamaning ildizlari x_1 va x_2 ga teng. Bunga ko'ra, $x_1^2 + x_2^2$ ifodaning qiymatini toping.

- A) $\frac{1}{6}$
- B) $\frac{1}{4}$
- C) $\frac{2}{2}$
- D) $\frac{1}{2}$

20. $f(x) = \cos^2 2x - \sin^2 2x$ funksiyaning eng kichik musbat davrini toping.

- A) $\frac{\pi}{2}$
- B) π
- C) $\frac{\pi}{4}$
- D) 2π

21. $y = f(x)$ funksiyaning minimum nuqtasi $(2k - 1; -3k)$, $y = -f(x + 4)$ funksiyaning maksimum nuqtasi $(k + 2; 2n - 1)$, $k \cdot n = ?$



- A) 66
- B) 55
- C) 77
- D) 84

22. Agar $f(x)$ funksiya uchun $c + xf(x) = (x - 3)(2x - 1)^5$ munosabat o'rinli bo'lsa, $f'(1)$ ni toping. (c – o'zgarmas son).

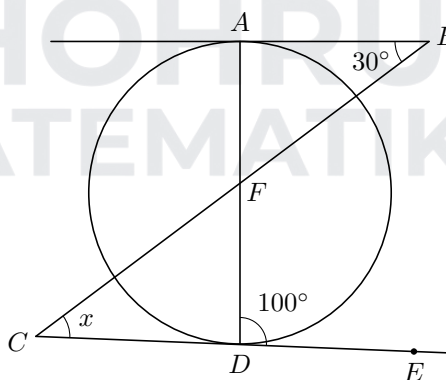
- A) -15
- B) -14
- C) -24
- D) -25

23. Integralni hisoblang:

$$\int_0^1 \arcsin x \, dx + \int_0^{\frac{\pi}{2}} \sin x \, dx .$$

- A) π
- B) 1
- C) $\frac{\pi}{2} + 1$
- D) $\frac{\pi}{2}$

24. BA va CD nurlar aylanaga mos ravishda A va D nuqtalarda urinadi. $\angle ABC = 30^\circ$ va $\angle ADE = 100^\circ$ berilgan. Yuqoridagi ma'lumotlarga ko'ra, $\angle BCD$ toping?



- A) 30°
- B) 35°
- C) 40°
- D) 50°

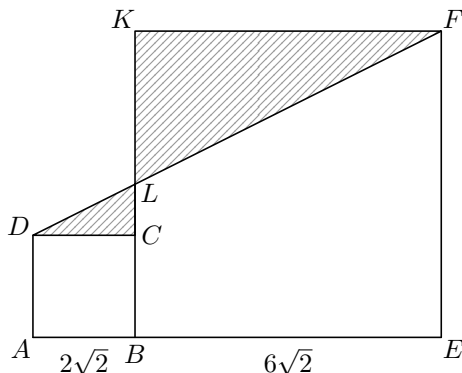
25. Aylananing O markazi to'g'ri burchakli ABC uchburchakning AC gepotenuzasida yotadi. Uchburchakning katetlari aylanaga urinadi. Agar OC kesmaning uzunligi 4 ga, C nuqtadan CB katetning aylana bilan urinish nuqtasigacha bo'lgan masofa 3 ga teng bo'lsa, CB ni toping.

- A) 7
- B) 8

C) $3 + \sqrt{7}$

D) $4 + \sqrt{7}$

26. $ABCD$ va $BEFK$ kvadratlar berilgan, bunda $AB = 2\sqrt{2}$ cm va $BE = 6\sqrt{2}$ cm. Shtrixlangan soha yuzini toping.



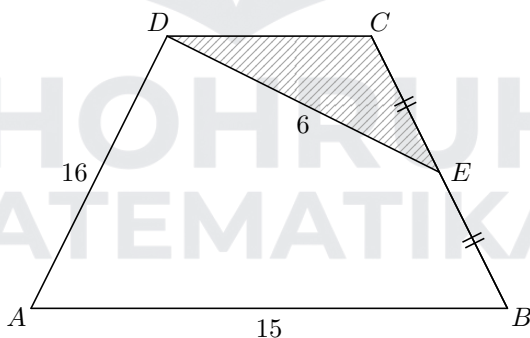
A) 18

B) 20

C) 15

D) 14

27. $ABCD$ – trapetsiya berilgan. Rasmdagi ma'lumotlarga ko'ra, $DC \parallel AB$, $ED \perp AD$, $CE = BE$, $DE = 6$, $AD = 16$ va $AB = 15$. U holda, shtrixlangan yuzani toping.



A) 12

B) 18

C) 16

D) 14

28. $ABCDEF$ muntazam oltiburchakda AC , CE , BF , FD diagonallar o'tkazilgan. AC va BF diagonallar L nuqtada, CE va FD diagonallar K nuqtada kesishadi. Agar oltiburchak tomoni $2\sqrt{3}$ ga teng bo'lsa, LK kesma uzunligini toping.

A) 4

B) 5

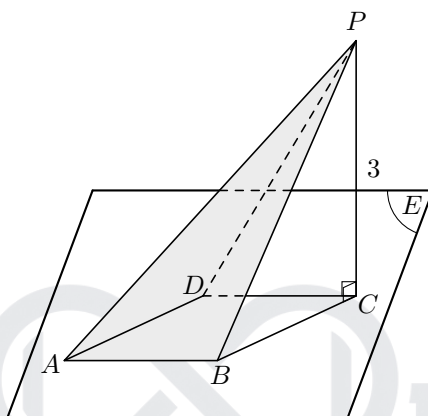
C) 6

D) 9

29. $ABCDEF$ muntazam oltiburchak. Agar $\overrightarrow{AB} = \vec{a}$ va $\overrightarrow{BC} = \vec{b}$ bo'lsa, $\overrightarrow{CD}$ vektori nimaga teng?

- A) $\vec{a} + \vec{b}$
- B) $\vec{b} - \vec{a}$
- C) $\vec{a} - \vec{b}$
- D) Hech biri

30. $ABCD$ kvadrat, $AB = 4$ sm, $CP = 3$ sm. Shaklda CP kesmasi $ABCD$ kvadrati joylashgan E tekisligiga perpendikulyar bo'lsa, PAB uchburchakning yuzini toping.



- A) 6
- B) 8
- C) 10
- D) 12

31. $A = \{1, 2, 3, 4, 5\}$, $B = \{2, 3, 4, 6, 8\}$ va $C = \{0, 3, 5, 7\}$ to'plamlar berilgan. $(A \cap B) \cup (A \cap C)$ ifodaning qiymatini toping.

- A) $\{2, 3\}$
- B) $\{3, 5\}$
- C) $\{2, 3, 5\}$
- D) $\{2, 3, 4, 5\}$

32. Buxorodan boshqa shaharlarga parvozlari bo'lgan ikkita aviakompaniya mavjud. Bu kompaniyalardan:

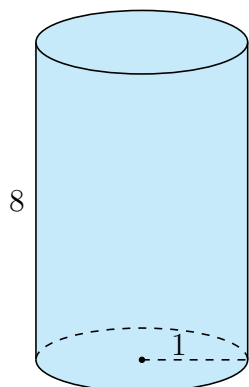
- biri Samarqandga kuniga 5 ta parvoz, Toshkentga kuniga 1 ta parvoz;
- boshqasi Samarqandga kuniga 4 ta parvoz, Navoiyga kuniga 2 ta parvoz

tashkil etmoqda. Bir kunda Buxorodan uchayotgan bu ikki kompaniyadan tasodifiy bittadan parvoz tanlab olinadi. Shunga ko'ra, tanlangan bu parvozlarning turli shaharlarga bo'lish ehtimoli qancha?

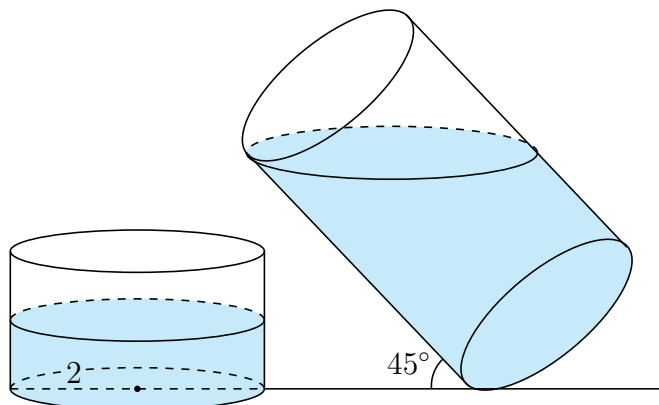
- A) $\frac{1}{2}$
- B) $\frac{2}{3}$
- C) $\frac{4}{9}$
- D) $\frac{5}{12}$

Topshiriqlar (33–35) va javob variant (A–F) larni o‘zaro moslashtiring.

Asosining radiusi 1 dm, balandligi 8 dm bo‘lgan to‘g‘ri silindr shaklidagi idish to‘liq suv bilan to‘ldirilgan (1-rasm). Suv to‘la idish 45° burchak ostida og‘dirilganda, undan ma‘lum miqdordagi suv asosining radiusi 2 dm bo‘lgan yetarlicha baland bo‘sh silindr idishga quyildi (2-rasm).



1-rasm



2-rasm

Javoblar.

- A) 2, 5
- B) 2π
- C) π
- D) 3
- E) 8π
- F) 4π

33. Idishda dastlab to‘la holatida necha l suv bor edi?

34. Kichik idishdagi suv balandligi necha cm ga teng?

35. Kichik idishga quyilgan suv hajmini toping? (l)

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36. Quyidagi parametrik tenglama berilgan:

$$(x^2 - x)a^2 + (x^4 - x^3 - x + 1)a - x^3 + x^2 = 0 .$$

a) Tenglama ikkita turli ildizga ega bo'ladigan barcha a parametrning qiymatlari sonini toping.

Javob: _____

b) Tenglama ikkita turli ildizga ega bo'ladigan barcha a parametrning qiymatlari yig'indisini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

37. Quyidagi tenglama berilgan:

$$\cos 5x - \sin 5x = \sin 7x - \cos 7x.$$

a) Tenglamaning $\left[0; \frac{\pi}{2}\right]$ kesmadagi eng katta ildizini toping? .

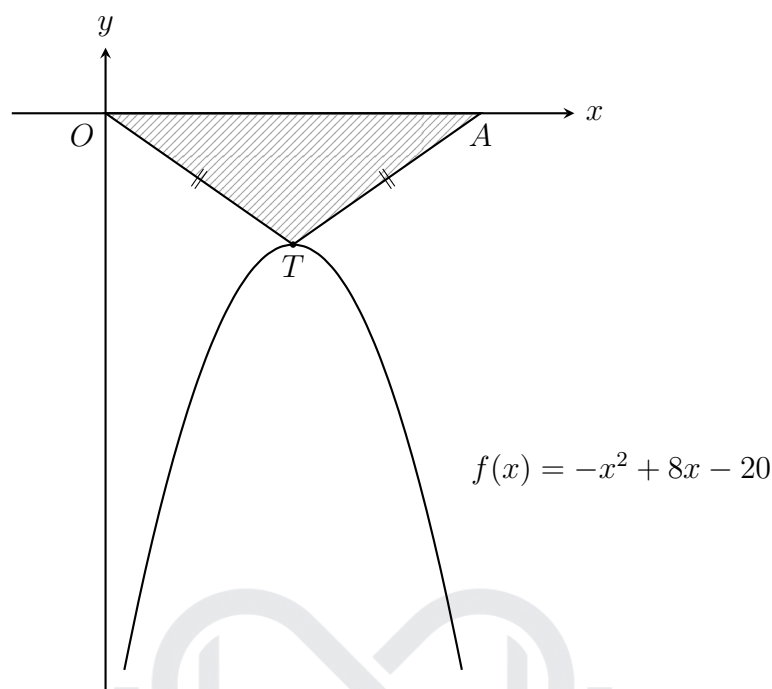
Javob: _____

b) Tenglamaning $x \in [0; \pi]$ kesmadagi ildizlar sonini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ko'chirib yozing.

38. Quyida uchi T nuqtada bo'lgan $f(x)$ funksiyaning grafigi berilgan. Agar $|OT| = |AT|$ ekanligi ma'lum bo'lsa,



a) OTA uchburchakning yuzini toping.

Javob: _____

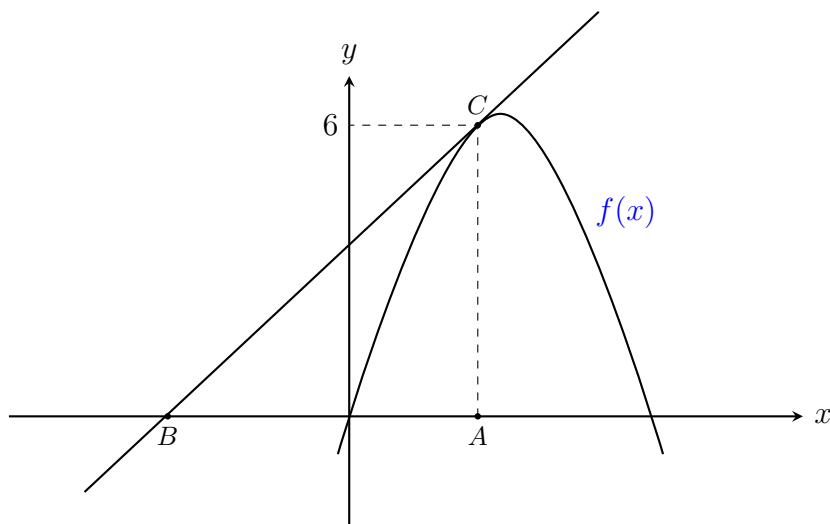
b) AT kema uzunligini toping.

Javob: _____

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39. Quyidagi shaklda d to'g'ri chiziq C nuqtada, $f(x) = -x^2 + 5x$ parabolaga urinma bo'lsa,



a) $|AB| + |BC|$ ni toping.

Javob: _____

b) $h(x) = (x^2 + 3x - 4) \cdot f(x - 2)$ bo'lsa, $h'(4)$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

40. $f(x)$ funksiya uchun $f''(x) = 18x^2 + 4x$, $f'(1) = 11$ va $f(1) = 6$ tengliklar o'rinli bo'lsin.

a) $f(0)$ ni hisoblang.

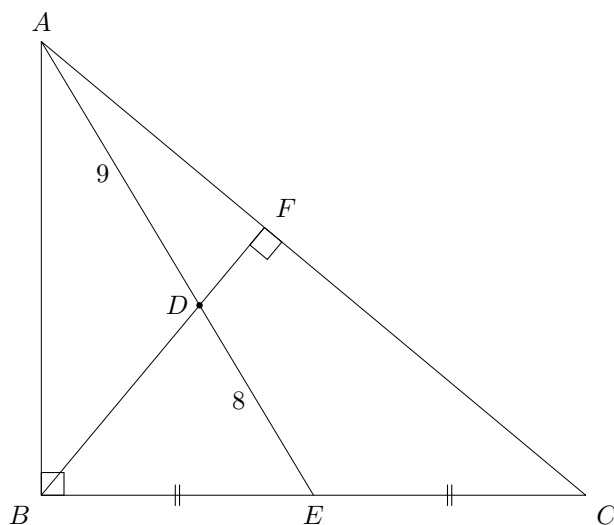
Javob: _____

b) $f(2026) = 5$ va $f(2025) = 2$ bo'lsa, $\int_{2025}^{2026} (f(x) + xf'(x)) dx$ ni hisoblang.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

41. ABC to'g'ri burchakli uchburchak berilgan. B to'g'ri burchakdan AC tomonga BF balandlik hamda A nuqtadan BC tomonga AE mediana tushurilgan. Ular D nuqtada kesildi. Agar $AD = 9$ va $DE = 8$ shart o'rinli bo'lsa,



- a) ABC uchburchak yuzini toping.

Javob: _____

- b) $|AC|$ gipotenuza uzunligini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiqa ko'chirib yozing.

42. $ABCD$ kvadratning DC tomonidan F nuqta, AF kesmadan esa shunday E nuqta olindiki, natijada AF va BE kesmalar kvadrat yuzasini teng uch qismga ajratdi. Bunda $EF = \frac{13}{3}$.

- a) Kvadrat yuzasini hisoblang.

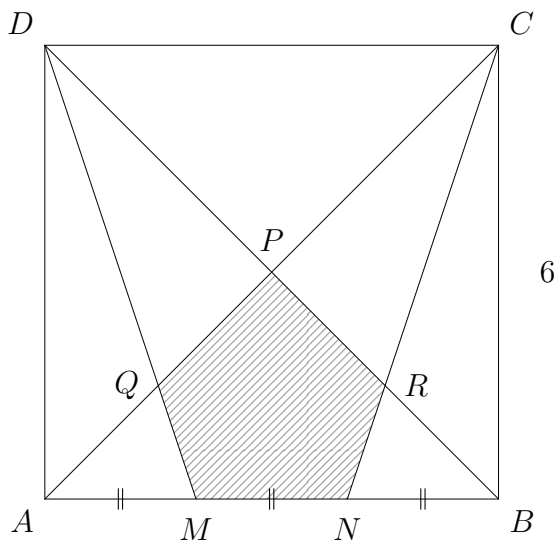
Javob: _____

- b) E nuqtadan BC tomongacha bo'lgan eng qisqa masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiqa ko'chirib yozing.

43. $ABCD$ kvadrat berilgan. AB tomondan uni teng uchga bo'luvchi M va N nuqtalar olingan. Rasmda ko'rsatilgandek $MQPRN$ beshburchak hosil qilingan bo'lsa,



a) $MQPRN$ beshburchak yuzini toping.

Javob: _____

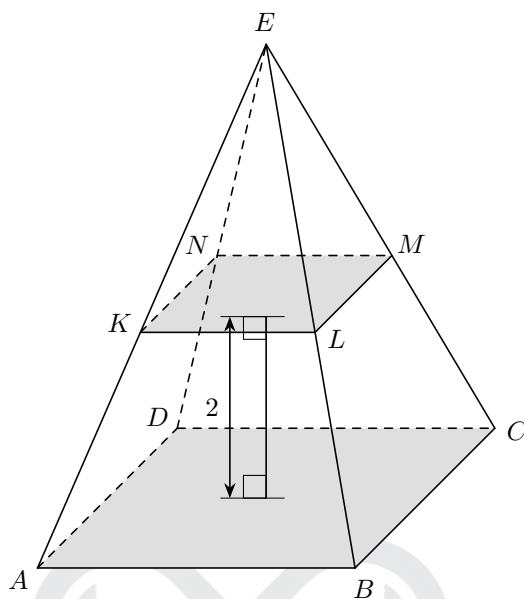
b) DPQ uchburchak yuzini hisoblang.

Javob: _____

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44. Shakldagi asosi kvadrat bo'lgan piramida berilgan. Asos tekisligiga parallel $KLMN$ tekislik bilan kesilgan. $S_{KLMN} = 9, \text{ cm}^2$, $S_{ABCD} = 16 \text{ cm}^2$ va asoslar orasidagi masofa 2 cmga teng bo'lsa,



- a) $EKLMN$ piramidaning hajmini toping. (cm^3)

Javob: _____

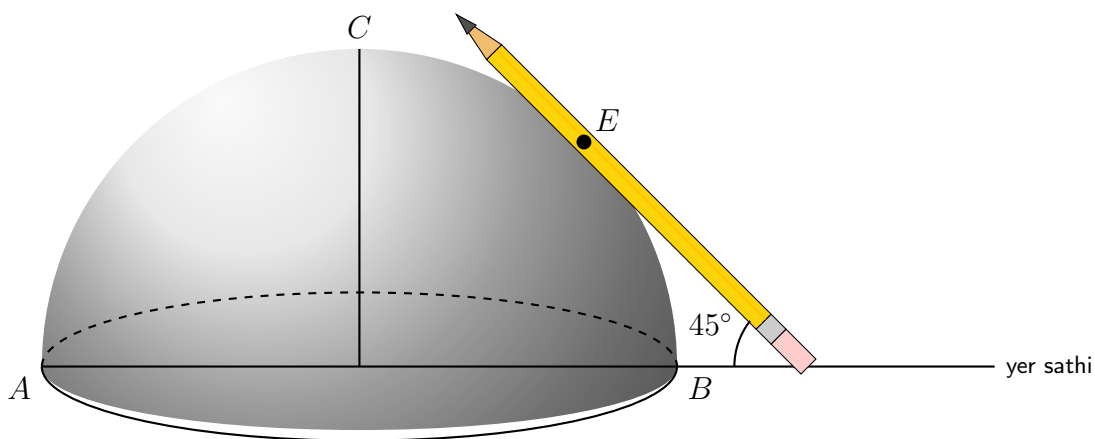
- b) $ABCDNMLK$ kesik piramidaga tashqi chizilgan shar to'la sirtini toping. (cm^2)

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

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45. Quyidagi rasmda kesimi parabolik bo'lgan shisha idishning A va B nuqtalari orasidagi masofa 12 cm va idishning eng yuqori nuqtasining yerdan balandligi 9 cm ga teng.



Bir qalam E nuqtada urinma bo'ladigan shaklda idish ustiga tushganda yer bilan 45° li burchak tashkil qiladi.

a) Bunga ko'ra, E nuqtaning yerdan balandligi necha cm?

Javob: _____

b) A va E nuqtalar orasidagi masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

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